
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

Anonymous authors
Anonymous submission

ABSTRACT

Algorithms are now the dominant information mediators of modern life – from social media feeds and targeted advertising to the rapidly expanding reach of LLMs. This creates a joint process in which agents fuse private, algorithmically-mediated signals with observational learning from their peers to reach decisions. We introduce a model of controlled sequential social learning that captures exactly this dynamic: an information-mediating planner (e.g., an LLM) sets the information structure faced by agents who simultaneously learn from the decisions of those before them. The planner may act to improve social welfare (an altruistic planner) or to induce a specific action it prefers (a biased planner). Our framework defines a new optimization problem for social learning, uniting dynamic programming with decentralized action choices and Bayesian belief updates. Within this setting we prove that the value function is convex and give a complete characterization of the optimal policies of both altruistic and biased planners, pinning down precisely how each trades the costs it incurs against the payoffs it earns from induced agent choices. This characterization shows that the optimal planner operates in sharply distinct modes determined by the public belief: investing the maximum allowed resource, investing nothing, or increasing or decreasing its investment as belief grows. Strikingly, over an identifiable range of beliefs the biased planner deliberately obfuscates the agents’ signals. Even under the most stringent transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—we prove that information mediation can move social welfare substantially in either direction. We corroborate the theory with simulations in which LLMs act as both planner and agents. The LLM-based planner reproduces the emergent strategic behavior our theory predicts, steering public opinion along the trajectories we derive; the deviations that remain are systematic and trace directly to non-Bayesian reasoning—the very cognitive patterns exhibited by human users and by LLMs trained on human-like data. Together, these results establish our framework as a tractable and behaviorally faithful foundation for studying, and regulating, the LLM information mediators now shaping public opinion.

1 INTRODUCTION

Large Language Models (LLMs) are now deployed as information mediators across socially-critical functions, from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants dispensing medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). This power cuts both ways: LLMs have repeatedly been shown to be persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) in their interactions with users. Crucially, at scale this algorithmic influence never operates in a vacuum. It collides with the organic spread of information as people observe and learn from one another, producing social dynamics that can sharply amplify or blunt a model’s impact—dynamics that any serious account of LLM influence must confront.

To make this concrete, consider an LLM-powered recommendation system that suggests a new restaurant to a sequence of potential customers. The quality of the restaurant—the true state of the world—is either good or bad, but this is unknown to the users and the system alike. Each user

sees a personalized recommendation (a private signal) and then decides whether to patronize the restaurant. The recommendation system (the *planner*) can invest resources to make its signal more informative; for example, by having the LLM generate a highly tailored review based on the user’s known preferences. This investment represents a realistic cost, corresponding to expenditures on market research, user surveys, or data acquisition needed to tailor information effectively.

A user’s decision, however, is never based on this private signal alone. Users also observe the actions of their predecessors—if the last ten users decided to patronize the restaurant, the next user will infer that they likely received positive information. This phenomenon, *social learning*, is a powerful force shaping human opinion across settings as varied as the adoption of vaccines (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), new technology (e.g. Gillingham & Bollinger (2021); Weber (2012)), and political and moral opinions (e.g., Brady et al. (2021); Guilbeault et al. (2018)). This paper’s central contribution is to open up the strategic interaction between a planner who controls private information and a population of agents who engage in social learning. A planner in this setting must anticipate how each of its actions not only influences the current user but also creates an information *externality* that reshapes the public belief confronting every future user.

Real-world interactions are immensely complex, but we distill this dynamic into a tractable formal model of *controlled social learning*, in which an algorithmic planner strategically chooses the precision of private signals for a sequence of agents at some cost. The planner’s lever is deliberately subtle: it never falsifies information, and only decides how much to invest in making its signals clear and informative. This new paradigm raises three critical questions, and our model lets us answer each of them decisively: How does a planner wield its power to steer collective beliefs? How should its strategy adapt to evolving public opinion? And what is the ultimate impact on social welfare?

The answers depend critically on the planner’s objective. An *altruistic* planner, like an ideal recommendation system, seeks to maximize user utility by helping them make the correct choice (patronize the restaurant if and only if it is good). In contrast, a *biased* planner, perhaps receiving a kickback from the restaurant, wants to induce a specific action (patronize) regardless of the true state.

This distinction is related to, but sharper than, the notion of alignment. An altruistic planner’s objective is, by definition, aligned with maximizing the agent’s *expected utility* under state uncertainty. A biased planner’s objective, being state-independent, is not—yet its actions are not always detrimental. When the true state happens to favor the planner’s preferred action (e.g., the restaurant is indeed good), its influence coincides with the agent’s *realized utility* in that instance. We map the full spectrum of these interactions, treating both altruistic and biased planners across scenarios in which their goals align with, or diverge from, those of the user.

Contributions and Outline This paper makes the following principal contributions:

1. **A Novel Theoretical Framework for Controlled Social Learning.** We introduce the first formal model to integrate a dynamic control problem for a centralized information planner with the mechanism of sequential social learning. The planner strategically chooses the precision of agents’ private signals to pursue an objective that is either altruistic (maximizing social welfare) or biased (inducing a specific action).
2. **A Complete Characterization of Optimal Planner Policies.** We fully characterize the optimal policies of both altruistic and biased planners as a function of the evolving public belief. For the altruistic case, the characterization rests on a new proof of the value function’s convexity that is of independent interest. These results lay bare the strategic trade-offs a planner faces when its actions impose informational externalities on future agents.
3. **Empirical Validation and Strategic Analysis Using LLMs.** We run simulations in which LLMs act as both planner and agents, and they establish three findings. (a) A planner that accounts for social learning influences public opinion and social welfare dramatically—far beyond what a myopic planner achieves. (b) The strategic behavior that emerges from the LLM planner closely tracks our theoretical predictions, demonstrating that the model is robust to non-Bayesian agent behavior. (c) LLMs display sophisticated strategic reasoning, underscoring both their potential and the societal risks they carry.

We position our work against the literature in Section 2, introduce our formal model in Section 3, and derive the optimal policies in Sections 4 and 5. We then put our findings to the test with LLM-

based simulations in Section 6 and conclude in Section 7. Proofs and additional results appear in the appendices.

2 RELATED WORK

Our work draws on and advances four literatures: social learning, information design, online persuasion and RL, and LLMs.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the sequential social learning framework in economics, and subsequent work has thoroughly characterized social-learning dynamics in more general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has also studied many variations in the kind of social information available to agents and various biases and misperceptions of the agents (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein).

The literature on the *control* of social learning remains sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The two closest works to our own, Wei & Anastasopoulos (2022) and Smith et al. (2021), both make assumptions we dispense with. Wei & Anastasopoulos (2022) requires two-way communication between the planner and agents (i.e., agents report their private signals to the planner in exchange for action recommendations). Smith et al. (2021) studies a benevolent planner who directly alters the choice rules of agents and derives a decentralized pivot mechanism to improve learning. Our model assumes neither two-way communication nor any control over the agents’ actions, which the agents retain in full. This makes it the appropriate model for information-mediating algorithms, which are typically invisible or black-boxed to the users they influence.

Information Design In our setting, the planner optimizes its objective by choosing the agents’ private signal precision—a constrained instance of the classical information design problem (Bergemann & Morris, 2019). Such models came into prominence with Kamenica & Gentzkow (2011)’s Bayesian persuasion framework, in which a planner chooses the signal structure informing a rational, Bayesian agent about an unknown state. Our planner solves a costly, constrained information design problem in every period, selecting a binary, symmetric signal at a cost. Against the many variants of information design already studied, our decisive departure is to embed social learning among the agents.

Only a handful of works consider information design in sequential social learning settings (Arieli et al., 2022; Wu et al., 2025), and both are confined to a one-shot setting in which the sender fixes a single information structure at the outset for the entire sequence of agents. Arieli et al. (2022) studies a self-interested sender who chooses a general information structure as in classic Bayesian Persuasion. Wu et al. (2025) constrain the information structure to be more informative in the sense of Blackwell and maximize the probability of a correct cascade. We break from both by solving a fully dynamic problem in which the sender chooses a fresh structure for each agent in the sequence, giving rise to a substantially richer class of sender strategies.

Online Persuasion and RL There is growing recent work at the interface of online learning and Bayesian persuasion/information design. These works study online sequential persuasion under unknown environment parameters, in which a sender interacts repeatedly with short-lived receivers while lacking knowledge of the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related set of papers examines sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or foresight by the receiver; these models emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across these works, the focus lies in learning-based performance guarantees such as sublinear regret relative to the best signaling scheme in hindsight and/or approximation algorithms under computational constraints.

Our model departs from all of these. The planner never learns unknown parameters; sequential dependence arises purely from social learning among receivers, not from the sender updating beliefs about the environment. Nor is the planner informationally advantaged: it observes neither the underlying state nor the private signals. Despite these constraints, we characterize the planner’s optimal policies exactly. Our setting is sequential but is emphatically not an online-learning or RL environ-

ment: actions never modify the underlying payoff-relevant state (there are no reward externalities), and the planner learns no unknown environmental parameters. The planner refines its knowledge solely by observing agents' actions, which reflect social learning and belief propagation rather than payoff exploration or state evolution. Sequential dependence thus comes entirely from information externalities, as each agent's action shapes the beliefs of those who follow.

LLMs A recent line of work studies LLMs specifically in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). Every one of these works is confined to one-on-one interactions and therefore cannot capture the social dynamics at the heart of our study. The closest is Duetting et al. (2025), who study information design in which a sender chooses both a signal structure and a 'framing' that conditions the receiver's (possibly non-Bayesian) posterior belief, and who deploy an LLM-based oracle to tackle the computationally hard task of optimizing over the framing space—building on the observation that LLMs can stand in for humans as economic agents (Dillion et al., 2023; Horton, 2023). Their work does not touch our central concerns of repeated interactions and social learning; we do, however, adapt parts of their methodology for our empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner and a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature determines a fixed, *unknown* exogenous state of the world $\omega \in \Omega := \{G, B\}$, where $\mathbb{P}(\omega = G) = b_1$, and b_1 is known to everyone. In the recommendation system example of Section 1, the state G corresponds to the case where the restaurant is good.

At each time i , the planner decides whether or not to invest in personalization for agent i . Subsequently, it provides agent i with a private signal $s_i \in \Omega$ to aid in decision-making. In our model, s_i is assumed binary and it matches the state of the world ω with probability $q_i \in [0.5, 1]$ i.e. $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i 's signal precision.

Each agent's signal is independent of those of other agents and the history when conditioned on ω . Based on this signal and the observed actions of previous agents $j < i$, agent i selects an action $a_i \in \Omega$. The agent receives utility 0 if her action matches the true state ($a_i = \omega$), and utility $-C$ otherwise, where $C > 0$. Each player chooses her action to maximize her own expected utility. For example, this corresponds to patronizing a business if and only if the state is G .

3.1 AGENTS' DECISION PROBLEMS

Each agent observes the actions of all her predecessors and their respective signal precisions. This history is denoted $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because all agents and the planner have identical information $\mathcal{H}_i$, there is a shared *public* belief about ω , which is updated after each agent acts. The public belief b_i just before agent i acts is $\mathbb{P}(\omega = G | \mathcal{H}_i)$, that before any agent acts is the a priori distribution over Ω , b_1 . As in the classic model of Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for the history $\mathcal{H}_i$ (see Appendix B.3 for proof).

Informed by $\mathcal{H}_i$ (equivalently, by the Markov property, b_i) and q_i , agent i chooses action $a_i \in \Omega$ so as to maximize her utility. If both actions fetch the same utility, she chooses the action that matches s_i . Agent i obtains a *private* belief $\tilde{b}_i$ about ω using $\mathcal{H}_i$ and her private signal s_i of precision q_i as follows (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

Using this private belief, she chooses the action corresponding to the state of the world that is more likely as per her posterior belief $\tilde{b}_i$. In other words, she chooses $a_i = G$ if $\tilde{b}_i > 0.5$; $a_i = B$ if $\tilde{b}_i < 0.5$; and $a_i = s_i$ if $\tilde{b}_i = 0.5$. Substituting for $\tilde{b}_i$, we find:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is then observed by subsequent agents and incorporated into the updated public belief b_{i+1} as follows (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly reveals her private signal via Equation (2). Thus, the updated public belief is identical to the private belief of agent i . Otherwise, from Equation (2), agent i 's private signal is uninformative and has no effect on her action. Thus, the public belief is unchanged, and an absorbing state, referred to as information cascade or herding in classic social learning literature such as Bikhchandani et al. (1992); Banerjee (1992), is reached. At such points, society (public belief) stop learning from agents' private information.

Agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$ which has the following form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We consider two types of planners: (1) an altruistic planner who wishes to induce agents to take the correct action ($a_i = \omega$) and (2) a biased planner who wishes to induce a specific action, say G , regardless of ω . We denote these different planners with subscripts A and B , respectively. In each case, the planner determines the precision of the private signal of each agent. A function $\beta(\cdot)$, which is non-negative, increasing, continuous, and concave in its argument, will denote the cost associated with the chosen precisions. The planner has an information set $\mathcal{H}_i$ identical to those of the agents.

Altruistic Planner At time i , the altruistic planner chooses the precision q_i for agent i and incurs a cost of $\beta(q_i)$ where $\beta(p) = 0, p \in [0.5, 1)$. For the altruistic planner, decreasing precision is never beneficial (see Appendix C.9). Thus, to simplify notation, the planner incurs additional cost only if it increases the precision above a baseline value of p , and the additional cost increases with further increase in the precision, with decreasing marginal costs. The agents know p and the function $\beta(\cdot)$.

The altruistic planner's goal is unambiguous: maximize social welfare—the expected total utility of the agents—net of the cost of precision investment. Let $r_A(b_i, q_i)$ be the instantaneous reward of the altruistic planner beginning at public belief b_i and choosing signal precision q_i for agent i . Starting from a public belief b_1 and following a sequence of policies $\pi = (\pi_i)_{i=1}^{\infty}$ such that $\pi_i(\mathcal{H}_i) = q_i$, the planner attains the following expected total discounted utility, for a discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The optimal utility and policy of the altruistic planner are then defined as the supremum and arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$ where Π is the set of all policies. From this formulation, the myopic value and policy are recovered by setting $\delta = 0$.

Biased Planner The biased planner's problem differs from the altruistic one solely in its objective, and hence in its cost and reward functions. (The political-campaign example in Appendix A makes the biased planner concrete.) The biased planner seeks to induce action G from every agent regardless of ω . When an agent chooses action G (respectively, B), the planner incurs cost 0 (respectively, $C > 0$), regardless of ω . The biased planner can make a private signal either more or less precise than the baseline value of p , and both directions incur costs. Any choice of precision other than p incurs a cost for the biased planner as it requires him to tailor the ad to an agent, which in turn needs research on how the agent best understands any content. The biased planner incurs cost $\beta(|q_i - p|)$ for choosing signal precision q_i , with $\beta(0) = 0$.

The biased planner's expected instantaneous reward at time i is then defined as follows:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ can now be defined for the biased planner, as $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ were defined for the altruistic planner using Equation (5).

As noted in the first paragraph of Section 3, each agent always earns higher utility by choosing an action that matches ω . Consequently, when $\omega = B$, the biased planner's success with an agent strictly lowers that agent's utility. The altruistic planner, by contrast, seeks to have every agent's action match ω whatever ω is, so its success always raises the agent's utility. The altruistic planner's objective is therefore aligned with each agent's at all times, whereas the biased planner's aligns only when $\omega = G$.

Both planners' utility-maximization problems are infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function given by Equation (3) (Puterman, 1990). A unique optimal value function therefore exists (Kallenberg, 2011). We restrict attention to deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *Our model rests on three assumptions, each of which deliberately handicaps the planner and thereby strengthens our conclusions: (1) The planner shares the agents' history. This binds the planner when it could hold richer data, but is innocuous when all parties rely on the same public signals, such as early findings or survey data. (2) The planner can reset agents' signal precision, but signals still pass through a binary symmetric channel. This binds the planner when it could cherry-pick, censor, or falsify signals, but is innocuous when its role is only to improve the quality of noisy data sources without altering content. (3) The planner's control choices are fully observable. This binds the planner when covert policies or framing are possible, but is innocuous in transparent environments. That the planner can still move welfare so dramatically despite these constraints is precisely what makes our findings striking.*

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic case, which isolates a planner that disregards social learning entirely. In our formulation this is exactly the choice $\delta = 0$: the planner ignores all future costs, and $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

Note that the myopic altruistic problem can equivalently be stated as a decentralized case in which each agent chooses the precision of her own private signal and incurs the associated cost with the goal of maximizing the sum of her own expected utility and cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimal policy therefore takes a clean threshold form: when public belief is sufficiently strong, the planner selects the baseline precision p at zero cost; when public belief is weak, it supplies a perfect signal, precision 1. The threshold depends only on the cost of the perfect signal ($\beta(1)$) and the cost of an incorrect action (C). Whenever $\beta(1) \geq C$, the perfect signal is simply too expensive relative to the cost of an incorrect action and is never used; here $t_M = 0.5$ and the belief interval calling for precision 1 vanishes entirely.

We now establish a foundational result for the altruistic optimal value function:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is substantial and, we believe, of independent interest. Its difficulty stems directly from the dependence of agents' actions on the public belief. Were the actions independent of the public belief process (as, e.g., in Nyarko (1994)), expected utility would be linear in the belief state and convexity would follow immediately; it is exactly the coupling we introduce that makes the result nontrivial.

This convexity is the key that unlocks a full characterization of the optimal policy.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & o.w. \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy splits cleanly into three phases in the public belief. First, exactly as in the myopic case, the optimal policy declines to invest in signal precision at extreme values of the public belief. But the optimal policy demands a *stronger* public belief before doing so than the myopic policy does—precisely because it, unlike the myopic policy, internalizes the effect on future agents.

Second, when public belief sits near 0.5 and carries almost no information, the optimal policy selects precision 1 whenever it is not cost-prohibitive. By Equation (2), the agent’s action then matches the true state of the world with probability 1, and the public belief collapses to either 0 or 1.

In the remaining phase, the optimal policy chooses the smallest precision $\max(b, 1 - b)$ that still lets the agent’s action reflect her private signal (see Equation (2)). Any lower precision would render the action uninformative beyond what agents already know. This is, precisely, the lowest-cost precision that keeps social learning alive through the observation of peers’ actions.

5 OPTIMAL BIASED POLICIES

As with the altruistic planner in Section 4, we begin with the myopic optimal policy. Paralleling Section 4, the myopic biased optimal policy $\pi_B^0(\cdot)$ is the arg supremum of the instantaneous reward $r_B(b, q)$. We now characterize $\pi_B^0(\cdot)$. A subtlety sets the biased case apart from the altruistic one: an optimal policy need not exist (Puterman, 1990, sec. 2.3.1). Where it does not, we pursue ϵ -optimal policies, marked with an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Proof in Appendix C.6. We now apply Theorem 4 to characterize the optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon*}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) reveals five sharply distinct phases, each with a clear strategic rationale.

When public belief is sufficiently strong (phase (A)), the cost of steering the system outweighs the benefit, even given the negative consequences for the planner’s utility when $b < 0.5$. The chosen precision p falls below $\max(b, 1 - b)$, so by Equation (2) the agent acts on the public belief regardless of her private signal. Because this action is uninformative, the public belief never moves; and since the policy is Markovian, the process repeats forever. This is an unfavorable cascade when $b < 0.5$ and a favorable one otherwise.

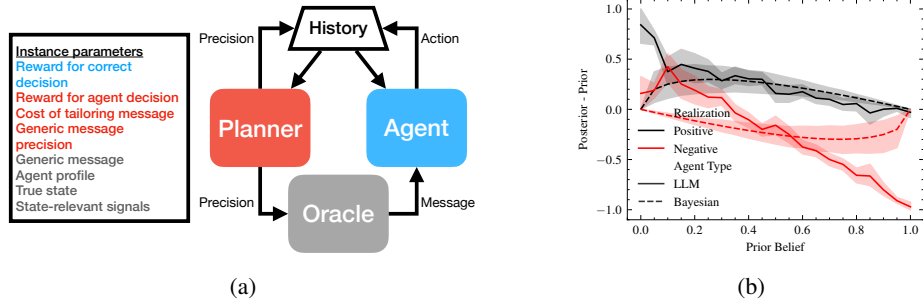


Figure 1: (a) A system diagram for our experiments. Instance parameters are color-coded by the LLM role that uses them. (b) The change in LLM (solid) and Bayesian (dashed) agents’ beliefs for a given prior after a positive (black) or negative (red) signal, making the non-Bayesian gap directly visible.

When public belief sits close to an unfavorable cascade (phase (B)), the planner raises signal precision high enough to sway the agent’s action, even though the resulting signal will, in expectation, be B . The agent then follows her private signal, which carries a nonzero chance of a favorable action for the planner. This is the planner’s calculated last-ditch effort to escape the unfavorable cascade.

For slightly higher belief (phase (C)), the planner *decreases* precision below p . Here $b < 0.5$, so more precise signals are more likely to deliver unfavorable news; the planner therefore holds precision just high enough to move the agent’s action ($q \geq 1 - b$)—the least precise signal that still offers a path toward a more favorable public belief.

When public belief weakly favors the planner’s desired action (phase (D)), the planner commits precision of at least p . Since $b > 0.5$, raising precision makes the agent more likely to infer $\omega = G$; investment here is the planner’s push to consolidate a favorable public belief.

When public belief is higher still (phase (E)), the planner deliberately lowers precision just below $\max(b, 1 - b)$ to $b - \epsilon$, so agents ignore their private signals and take action G . In this regime the risk that a private signal overturns the favorable public belief outweighs both the cost of lowering precision and the upside of pushing belief higher.

6 EVALUATION VIA LLM-BASED SIMULATION

We now put our model and theory to an empirical test with LLMs, probing both their implications and their reach. The evaluation proceeds in three steps: we first characterize the behavior of LLM agents to pin down their key deviations from Bayesian rationality, then show how an LLM planner adapts its strategy in response, and finally quantify the resulting impact on social welfare.

To realize the controlled social learning setting, we operationalize our model in a scenario in which agents decide whether to buy a new car. As shown in figure 1a, we employ LLMs in three roles, described below with corresponding prompts in Appendix E.2.2:

Agent: Role-playing as an assigned profile, agents observe previous actions and a private message about the car of known precision, form their belief about the car’s quality, and decide to buy or not.

Planner: After observing the history of actions, the planner selects the precision q_i of agent i ’s private signal according to their objective

Oracle: Given the choice of precision by the planner, an agent profile, and a fact sheet about the car (see Appendix E.2.1), the oracle generates a private signal of desired precision tailored to an agent. We benchmark the outcomes of these LLM-based simulations against numerical evaluations of the planner’s MDP. In both the LLM and numerical experiments, we adopt a linear cost function, $\beta(q) = k|q - p|$, and vary k , baseline precision p , and discount factor δ , holding the cost of an incorrect action C fixed at 1. Appendix E details the experimental setup and prompting, and Appendix E.3 validates both the agents’ beliefs and the oracle’s performance.

Our experiments establish three results:

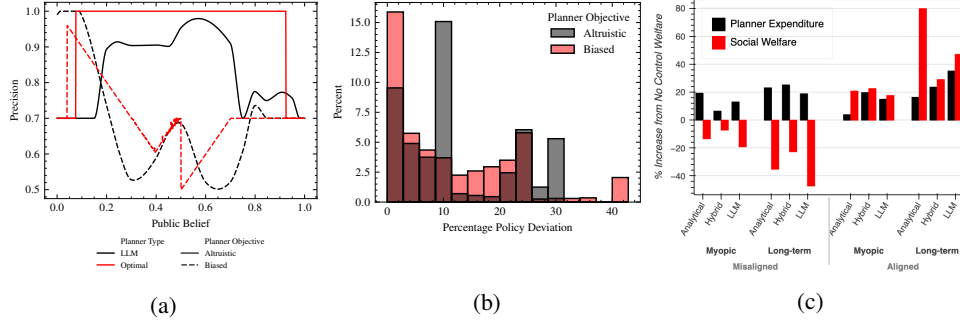


Figure 2: (a) Example policies from the LLM planner (black) and the analytically optimal planner (red) in altruistic (solid) and biased (dashed) settings, showing their close agreement. (b) The distribution of the percentage deviation between the LLM and optimal policies, concentrated near zero. (c) Planner expenditure and social welfare change as a percent of the no-control baseline welfare, with the true state fixed to B. The left half is a biased planner seeking action G; the right half is an altruistic planner.

1. **The importance and impact of social learning control:** Planners—both analytical and LLM—dramatically improve their own utility by accounting for and capitalizing on social learning dynamics. Depending on the alignment between planner and agent objectives, this either helps or harms social welfare.
2. **The robustness of our analytical characterization:** LLM planners choose policies strikingly close to the non-obvious analytically optimal policies, even while facing non-Bayesian agents played by other LLMs. This demonstrates that our characterization is robust well beyond its idealized assumptions.
3. **The emergent strategic behavior of LLMs:** LLM planners exhibit sophisticated emergent strategic behavior. Their policies not only track the optimal policies but also deviate systematically in exactly the ways needed to accommodate the non-Bayesian agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

Our analytical results assume Bayesian agents, yet both humans and LLMs are well documented to exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate the LLM’s biases, we measure directly how an LLM’s belief updates in response to new information. These measurements, shown in figure 1b, expose three clear non-Bayesian patterns in LLM belief updating (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.
- (NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

These same cognitive biases have been documented empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)), grounding our LLM agents in real human behavior. The sections that follow show that the LLM planner’s policy remains robust in the face of non-Bayesian agents of exactly this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now compare the policies of LLM planners against the optimal policies derived in Sections 4 and 5. As figure 2a shows, the emergent strategies of the LLM planner match the theoretical optimum with remarkable structural fidelity, in spite of the deviations in agent behavior. Even when the optimal policy is highly non-trivial (e.g., the biased policy in figure 2a), the LLM planner exhibits sophisticated strategic behavior that both accounts for and exploits social learning.

In the altruistic case, both planners halt investment when public belief is strong and invest heavily when it is uncertain. In the biased case, both follow the same qualitative playbook: heavy investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once a favorable belief takes hold. The magnitude of the policy deviation is small, as figure 2b makes clear: for both altruistic and biased planners, the deviation stays below 10% across the majority of belief states, confirming the tight structural alignment between the emergent LLM strategy and our theoretical characterization.

The structural differences that do remain are not noise—each is a precise strategic adaptation to a specific non-Bayesian behavior identified in Section 6.1. (1) The LLM planner avoids extreme precisions (0.5 or 1.0), consistent with a known central-tendency bias (Rupprecht et al., 2025). (2) The LLM planner tapers investment gradually rather than cutting it off sharply. This is a direct response to the agents’ resistance to cascades (NB3): the planner learns that it is never entirely “safe” to stop investing, since even agents with strong priors can be swayed by a countervailing private signal. (3) The biased LLM planner keeps investing at very low beliefs. This reflects a grasp that its agents may overreact (NB2) to a surprisingly positive signal, making a “last-ditch effort” more viable than the Bayesian case would suggest.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never harms an agent’s expected utility, social welfare is monotonic in signal precision (see Appendix C.9); it follows that the altruistic planner always raises social welfare above the baseline. To quantify these effects, we compare social welfare and planner expenditure across three settings: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

figure 2c shows that planners in all settings move social welfare substantially, and that neglecting social learning (the myopic cases) sharply worsens the planner’s outcomes. Most strikingly, the biased analytical and LLM planners cut social welfare by 40 to 50% when misaligned—achieved entirely by intentionally obscuring information about the true state through policies that lower signal precision. That a planner can inflict damage of this magnitude is all the more remarkable given the harsh constraints it operates under (see Remark 2), and it makes the risk posed by misaligned LLM information mediators concrete rather than hypothetical.

The results also expose the real cost of a misspecified model of agents. In the hybrid setting, the analytically optimal policy—designed for Bayes-rational agents—proves brittle, and its performance degrades when confronted with non-Bayesian agents. The LLM policy, though structurally close to the analytical one, is far better calibrated to non-Bayesian agents carrying human-like biases.

7 CONCLUSION

We introduced a novel framework for the strategic control of social learning by an algorithmic information mediator and built a formal model of this increasingly consequential dynamic. We proved that even a tightly constrained planner can drastically move social welfare, for better or worse, and that accounting for social learning is essential to wielding this influence effectively. Our simulations bore out these insights: LLMs display strategic reasoning that mirrors the character of our analytical results while adapting to non-Bayesian agents.

Together, these findings yield three firm takeaways for algorithmic information mediation. (1) The impact of human-AI or human-algorithm interaction cannot be studied in a vacuum: ignoring its interplay with other interactions such as social learning means missing its true potential, positive or negative. (2) Information mediators wield immense power to guide or derail social learning and public opinion even through subtle influence, and this power demands sustained study of how to mitigate its risks. (3) Modern LLMs exhibit emergent strategic behavior that accounts for and exploits both social learning and non-Bayesian cognitive biases. The latter two points make plain the urgency of understanding how to mitigate the risks of algorithmic information mediators, and of LLMs in particular. The one clear limitation of our study is the absence of human data; because the fidelity of LLM-human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable for validating, and perhaps fine-tuning, the LLM simulators.

We close with two directions we regard as most promising: (1) generalizing our model and results, and (2) mitigating the negative welfare effects of biased planners. Several generalizations would relax our modeling assumptions and bring the framework closer to reality: larger state spaces, more general signal structures, more general loss functions, and heterogeneous agents. Moving beyond binary states and 0-1 losses is primarily an algebraic undertaking, and we are confident the qualitative character of our results will persist. For more general signal structures, the central challenge is preserving the convexity result of Theorem 2; identifying the broadest class of signal structures for which convexity holds is a particularly compelling open problem. As a first step toward heterogeneous agents, we already generalize the altruistic planner results to agents of opposing preferences in Appendix D.

A second important direction is preventing the welfare losses that biased planners inflict. This calls for regulations or mechanisms that align the incentives of planners and agents and thereby avert detrimental welfare outcomes. Charting the strategic dynamics, as we have done here, is the essential first step toward designing such safeguards.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbbed5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.
- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.
- Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.
- Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.
- Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.
- Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.
- Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.

594 Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change
595 as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

596 Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learn-
597 ing. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi:
598 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.

599 Ilai Bistriz, Nasimeh Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic
600 agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.

601 William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies
602 moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.

603 Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive
604 effort and moral-emotional language of LLM arguments, April 2024. URL [http://arxiv.org/abs/](http://arxiv.org/abs/2404.09329)
605 2404.09329. arXiv:2404.09329 [cs].

606 Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persua-
607 sion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Cur-
608 ran Associates, Inc., 2020. URL [https://proceedings.neurips.cc/paper/2020/hash/](https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html)
609 ba5451d3c91a0f982f103cdbc249bc78-Abstract.html.

610 Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Per-
611 suasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR,
612 July 2021.

613 Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updat-
614 ing, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].

615 Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.

616 Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.

617 Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants?
618 *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.
619 tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0)
620 S1364-6613(23)00098-0. Publisher: Elsevier.

621 Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song
622 Zuo. Information Design With Large Language Models, September 2025. URL [http://arxiv.org/](http://arxiv.org/abs/2509.25565)
623 abs/2509.25565. arXiv:2509.25565 [cs].

624 Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J.
625 Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of*
626 *Machine Learning Research*, 18(26):1–5, 2017.

627 Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal:*
628 *Microeconomics*, 2:221–243, 11 2010.

629 Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative
630 News Recommendation. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, pp. 3444–3453,
631 New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi:
632 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.

633 Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Hu-
634 man Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>.
635 arXiv:2410.19599 [econ].

636 Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management*
637 *Science*, 67(11):7091–7112, 2021.

638 Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation
639 of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.

640 Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through
641 Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].

642 Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16):
643 1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL [https://doi.org/](https://doi.org/10.1001/jama.2023.5321)
644 10.1001/jama.2023.5321.

-
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.
- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.
- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.
- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klemp. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.

-
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weisser, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents in our terminology) deciding whether to patronize a business or service provider (e.g., a restaurant, contractor, or realtor) and a recommendation system (the planner) that serves the agents information about the business. The unknown binary state of the world is whether the business is good. Both the recommendation system and the agents want their choices to match this state: agents want to patronize the business if and only if it is good, and the recommendation system wants to induce exactly that. In this sense the recommendation system is an *altruistic planner*.

The recommendation system can show an agent a highly targeted (*precise*, in our terminology) ad that lays out the strengths and weaknesses of the business in contexts she relates to given her background, or it can show her a generic or confusing ad that does little to inform her action. The precise signal is more informative, comprehensible, and relatable, and so is more likely to drive her to the correct conclusion—patronize the business if and only if it is good. Social welfare rises as more agents reach correct decisions. But changing the precision of an ad incurs a cost, since it requires tailoring to the agent’s specific background. The recommendation system therefore chooses the precisions to maximize social welfare net of these costs.

We now give a real-life example of a *biased planner*. Consider a group of voters (agents) motivated to support the candidate most likely to win in their community. The pull to “back a winner” demonstrably shapes electoral outcomes—it is why US states with early primary elections exert an outsized influence on results (see Bartels (1988)). Here the planner is a specific candidate’s campaign, which seeks to motivate agents to back that candidate. The binary true state of the world is whether the candidate is winning, so an agent who knew the true state would back the candidate if and only if he were winning.

Each agent understands her community well and would know whether the candidate is winning (the true state) if she knew his stances and policies—which she would grasp correctly whenever they are conveyed in the form

she digests best. Some agents absorb audio-visuals best, some long-form articles backed by facts, figures, and citations, some only brief and focused content, and some only their native language.

The campaign sends digital ads (private signals) to agents at varying degrees of precision, where precision captures how closely the content is tailored to the agent’s taste. A precise signal conveys the true state accurately, helping the agent clearly understand the candidate’s policies, track record, and character so she can correctly infer whether he will win. Crucially, a precise signal does not guarantee that the agent backs the planner’s candidate: if the candidate is in fact losing in the agent’s community, a highly precise signal would drive her to oppose him. An imprecise signal, by contrast, is more likely to induce her to back him, because it obscures the true state (by being vague or confusing) and thereby raises the chance she believes he is winning. The biased planner is thus incentivized to *decrease* an agent’s signal precision precisely when it believes its candidate is losing. Tailoring the ad requires researching how the agent best understands content, which is costly for the planner. Even making the signal deliberately imprecise is costly, as it too requires agent-specific tailoring: to lower precision, the planner must know, say, that an agent best understands focused, brief messages so it can confuse her with long-form, verbose, detailed articles. The planner selects the precisions to maximize the expected number of backers net of the cost of generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of developments at step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Note that from (1), q_i is a function of $\mathcal{H}_i$. Thus, if a random variable is conditioned upon both q_i and $\mathcal{H}_i$, the conditioning on q_i is redundant. We will use this principle throughout.

B.1 PRIVATE BELIEF UPDATE

We now describe how agent i chooses action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ be the public likelihood ratio based on observed actions. Agent i forms a private belief about ω , $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$, by combining the current public belief with her private signal, s_i . We consider a private likelihood ratio, $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. We now describe how $\tilde{\alpha}_i$ and $\tilde{b}_i$ are computed.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In obtaining Equation (8), we apply the assumption that, conditioned upon ω , agents’ signals are independent of one another and the history to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same is done when conditioning upon $\omega = B$. We additionally use the fact that q_i is a function $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

We let $y(b, q)$ and $z(b, q)$ denote the probability of realizing signal G or B , with public belief b and signal precision q conditioned upon the history. By conditioning on ω and applying the Law of Total Probability,

these probabilities can be expressed as follows:

$$\begin{aligned}
y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\
&= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\
&= q_i b_i + (1 - q_i)(1 - b_i) = 1 + 2q_i b_i - q_i - b_i
\end{aligned} \tag{11}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \tag{12}$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1 - q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \tag{13}$$

Note that if $s_i = G$ then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot be 0. Similarly, if $s_i = B$, then $z(b_i, q_i)$ cannot be 0. Thus, the equation above is well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i will be G if $\tilde{b}_i$ exceeds 0.5 (equivalently, if $\tilde{\alpha}_i > 1$), B if $\tilde{b}_i < 0.5$ (equivalently, if $\tilde{\alpha}_i < 1$), and s_i otherwise as per the tie-breaking rule. This action and i 's precision q_i are observed by all peers, resulting in an updated public belief b_i .

When $b_i < 1 - q_i$, then $\alpha_i < \frac{1 - q_i}{q_i}$. Therefore, if $s_i = G$, $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, $\tilde{\alpha}_i < \frac{(1 - q_i)^2}{q_i^2} \leq 1$ (since $q_i \geq 0.5$) by Equation (10). Thus, no matter the private signal realization s_i , $a_i = B$ when $b_i < 1 - q_i$. Similar arguments can be used to show that when $b_i > q_i$, $a_i = G$. Intuitively these cases are when the strength of the public belief overpowers the precision of the private signal.

Now consider the case that $1 - q_i \leq b_i \leq q_i$. Then if $s_i = G$, $\tilde{\alpha}_i \geq 1$ from Equation (8) and since $b_i \geq 1 - q_i$. If $s_i = B$, then $\tilde{\alpha}_i \leq 1$ from Equation (10) and since $b_i \leq q_i$. Together with the tie-breaking rule, this results in action $a_i = s_i$. This leads to Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Thus, agent i 's action is only informative in the first case of Equation (2) where it allows future agents to perfectly infer i 's private signal. Thus, in this case, public belief will update just as the private belief from Equation (13). In the latter two cases of Equation (2), the action contains no information about the true state. Thus, the public belief will not change. This yields Equation (3) reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \tag{14}$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. It follows then that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \tag{15}$$

From Equation (2), for the second and third cases, it is clear that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned}
&\mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\
&= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\
&= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\
&\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\
&= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\
&\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\
&= b_i q_i + (1 - b_i)(1 - q_i)
\end{aligned} \tag{16}$$

The first step applies the law of total probability. In the second step, using the fact that q_i is a function of $\mathcal{H}_i$, we substitute b_i for $\mathbb{P}(\omega = w|\mathcal{H}_i, q_i)$. In the third step, we use the fact that, for $1 - q_i \leq b_i \leq q_i$, $a_i = s_i$, and, further, that s_i is independent of the history when conditioned upon ω . The result follows.

This leads to the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G|b_i, q_i, a_i\} \quad (17)$$

In other words, b_i captures the entire information pertinent to the update of b_{i+1} that is contained in $\mathcal{H}_i$.

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G|b_i, q_i\}}{\mathbb{P}\{a_i|b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

We have thus shown that the public belief is Markovian, adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992). These adaptations are necessary precisely because the central planner's control over the agents' precisions departs from the classic setting.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in [\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}]$. This yields Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove by induction that the expected k -th stage reward—the planner's expected utility from the control and action of the k -th agent—is convex. The instantaneous reward (Equation (5)), convex in public belief, furnishes the base case.

The first obstacle is the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update places support on at most 2 values. We handle this with a decision tree: the complete binary tree of all trajectories once an initial belief and policy are fixed. Each node has two children, one per possible signal realization the next agent might receive. The root is the expected instantaneous reward at time 1, from the first agent's action, and the induction descends the tree, with the k -th level holding 2^{k-1} nodes, each tied to a sequence of $k - 1$ signal realizations.

We then show that a $(k - 1)$ -th level node with convex expected reward passes that property to both of its k -th level children. This is exactly where the dependence of agents' actions on public belief must be confronted:

even under identical precision and an identical signal realization, two agents starting from different public beliefs can take opposing actions (see Equation (2)), so the standard tools for bounding future terms of a Markov process do not apply. Our specific belief update turns this difficulty into an asset. We exploit the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$), and together with the convexity of instantaneous rewards this closes the inductive step and completes the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

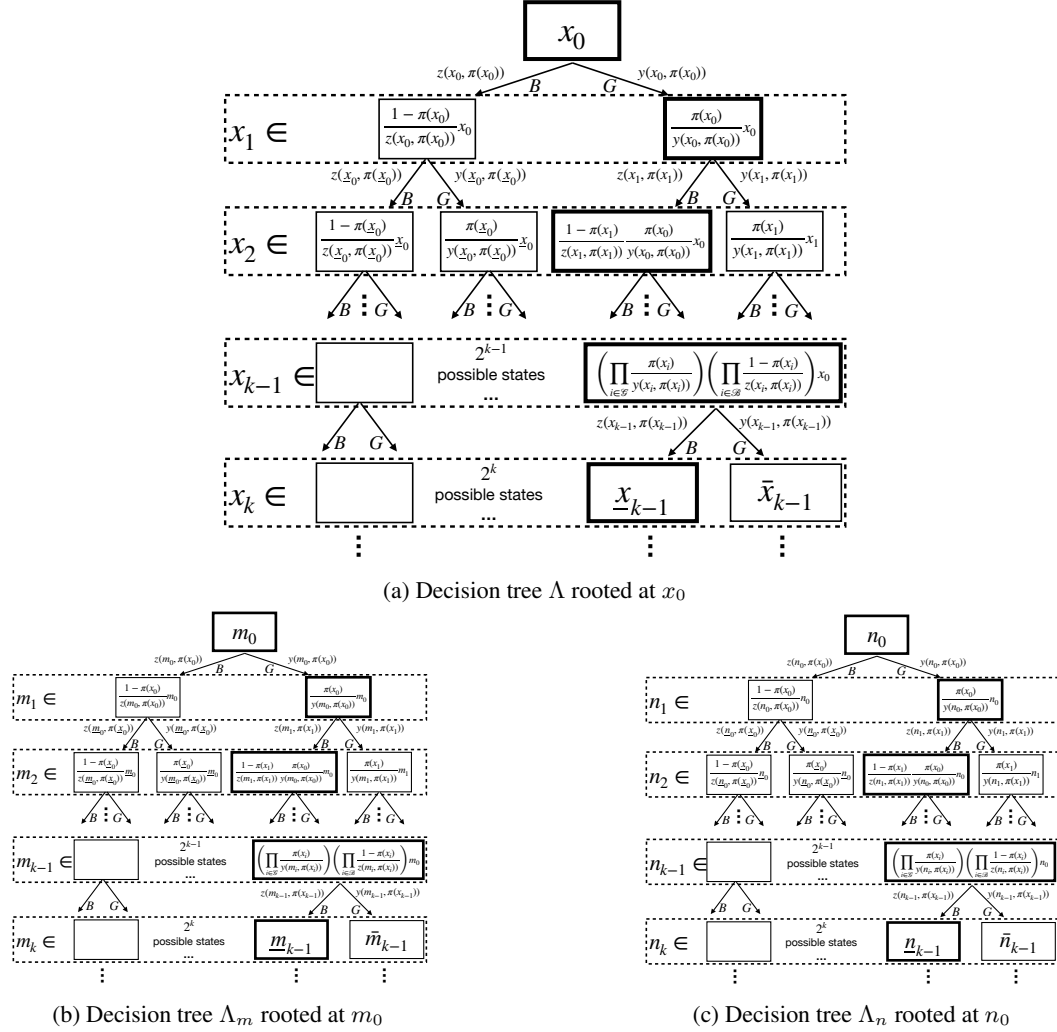


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon, 0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon, 0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We now show that, for the altruistic planner, the problem we consider is exactly equivalent to one involving two types of individuals with opposing preferences—a first, concrete step toward heterogeneous agents.

Assume that there are two types of agents G and B with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise. The second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further, assume that all agents know their own type and the types of all other agents. All other aspects of the model remain the same.

The private belief update of agent i can be expressed as below. This remains unchanged because it only depends on the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_iq_i-b_i-q_i} b_i & s_i = G \\ \frac{1-q}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows with superscript indicating the type of agent i and $!$ indicating negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because we assume the type of each agent is known to all, when $q_i \geq \max(b_i, 1 - b_i)$, all agents can perfectly infer the private signal of agent i . If $q_i < \max(b_i, 1 - b_i)$, the action of agent i is uninformative.

Thus, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$ as these choices align with the personal utility functions of the agents.

Thus, we can express the planner's instantaneous reward as follows with superscript denoting the type of agent i :

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

These two reward functions are identical to each other and to the reward function in Equation (5), which establishes that the altruistic planner's problem is unchanged in the presence of heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate the controlled social learning setting, we consider a sequence of agents deciding whether to buy a new car model, each informed by observations of past actions and an ad whose precision the planner selects. LLMs filled three distinct roles: (1) the agents, (2) the planner, and (3) an oracle. Each agent was assigned a ‘profile’ capturing demographic information along with car use cases and preferences. At the outset we fixed either the good or the bad car as the offering. The information sets and objectives of the planner and agents match the model of Section 3 exactly.

The oracle played two roles: (1) it generated a generic message serving as the baseline-precision signal absent any planner intervention, and (2) once the planner chose a precision, it generated a message at that precision, customized to the target agent’s profile.

The oracle received a fact sheet (safety rating, customer satisfaction, and so on) for each type of car along with the agents’ profiles. Together, the fact sheet and profile let it generate a tailored message at the planner’s prescribed precision. Appendix E.2 gives full detail on the setup and prompting.

Each agent made a binary decision from observations of previous decisions and its private signal, with the goal of matching the true parameter value.

The planner received the same decision history and chose a precision level for each agent’s message. It was also given the precision of a generic, un-tailored message that it could send for free. In the altruistic case the planner’s objective was to induce correct decisions; in the biased case, to induce agents to buy the car.

We benchmark the LLM simulations against the analytic solutions obtained via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments were run with the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations over the parameters in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Instance parameters used across all analytical and LLM experiments.

The car facts were designed so that the good car was unambiguously good for everyone and the bad car unambiguously bad for everyone. The fact sheets for the GOOD and BAD cars follow.

```
fact_sheet["GOOD"] = [  
    "EPA-rated range: 290 miles.",  
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",  
    "8-year / 100,000-mile battery warranty.",  
    "NHTSA 5-Star overall safety rating.",  
    "Cold-soak -5 °F keeps 93 %% range.",  
    "Cools cabin 100→72 °F in 3 min.",  
    "Cargo volume 26 ft3 total.",  
    "Tow's 1 500 lb with factory hitch.",  
    "Average 1.2 service visits first 3 yrs.",  
    "92 %% owners would buy again (Apr-25).",  
    "Qualifies for federal credit.",  
    "Quarterly OTA updates incl. battery pre-heat."  
]  
fact_sheet["BAD"] = [  
    "EPA-rated range: 155 mi (combined).",  
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",  
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
```

```

1836         "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
1837         "Cold-soak 15 °F retains 58 %% range;cabin heat cuts range 20%.",
1838         "Cabin cools 105 °F → 72 °F in 11 min.",
1839         "Cargo volume 17 ft3; rear seats fixed (no fold-flat).",
1840         "No factory towing approval.",
1841         "Average 3.8 service visits per year (battery-alert issues).",
1842         'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1843         "Not eligible for federal tax credit.",
1844         "OTA updates limited to infotainment."
1845     ]

```

Profiles were generated programmatically and included each agent's name, age, city of residence, interests, planned use case for the car (e.g., commuting or road trips), and major pain points or concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1852 sys = (
1853     f"You are deciding whether to buy a car. "
1854     f"Your profile is {profile}"
1855     f"You start believing there's a 50% chance the car is GOOD. "
1856     f"You have two information sources: "
1857     f"1. Previous agents' decisions "
1858     f"2. An advisor's message about the car "
1859     f"Your goal is to maximize your reward "
1860     f"i.e. BUY if the car is GOOD and "
1861     f"NOT BUY if the car is BAD. "
1862     f"If you BUY a GOOD car, you GAIN ${C}. "
1863     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1864     f"If you BUY a BAD car, you LOSE ${C}. "
1865     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1866     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1867     f"decision = exactly 'BUY' or 'NOT BUY'."
1868 )
1869 user = (
1870     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1871     f"ADVISOR MESSAGE:\n{message}\n\n"
1872     f"TASK: Use the previous decisions and the advisor message. "
1873     f"Decide BUY or NOT BUY. "
1874     f"Update your belief about car quality (0.0-1.0). "\n"
1875     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1876 )

```

Planner LLM Prompt:

```

1876 sys = (
1877     f"You are providing messages to potential car buyers. "
1878     f"Each potential car buyer (or agent) has the same information. "
1879     f"Your goal is to get agents to {objective}."
1880     f"Each decision according to your objective, earns you {C} dollars. "
1881     f"You and the agents have the same information. "
1882     f"Initially there is a 50% chance the car is GOOD. "
1883     f"The agents' aim to BUY if the car is GOOD "
1884     f"and NOT BUY if the car is BAD. "
1885     f"You can send a baseline message for free "
1886     f"or tailor the message to the agent at a cost. "
1887     f"Tailoring costs {k} dollars times the change in precision. "
1888     f"(Message precision is the probability (0.5 to 1.0) "
1889     f"that the message leaves the buyer with the correct impression. "
1890     f"For example, if the car is likely GOOD: "
1891     f"a high precision message will likely suggest BUY "
1892     f"and a low precision message will likely suggest NOT BUY. "
1893     f"You may choose to increase or decrease precision. "

```

```

1890     f"Agents decide based on your message and the decision history. "
1891     f"They update beliefs by observing previous decisions. "
1892     f"IMPORTANT: Thus your objective is to maximize:"
1893     f"{C} * (# of agents who choose according to your objective) "
1894     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1895     f"where {d} is your discount factor for future cost. "
1896 )
1897 user = (
1898     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1899     f"DECISION HISTORY:\n{history_txt}\n"
1900     f"TASK:"
1901     f"Based on the history and your objective, "
1902     f"choose your desired precision.\n"
1903     f'JSON response: {"message_precision": <number>}}\n'
1904 )

```

Oracle LLM Prompt:

```

1906 sys = (
1907     f"You will be given "
1908     f"1. an agent profile, "
1909     f"2. information about a car, "
1910     f"3. the true quality of the car, "
1911     f"4. a baseline message and the 'precision' of that message, and "
1912     f"5. a desired precision for the message you output. "
1913     f"(Message precision is the probability (0.5 to 1.0) "
1914     f"that the message leaves the buyer with the correct impression. "
1915     f"For example, if the car is likely GOOD: "
1916     f"a high precision message will likely suggest BUY "
1917     f"and a low precision message will likely suggest NOT BUY. "
1918     f"IMPORTANT: Output a message with the requested precision. "
1919     f"IMPORTANT: The message should be tailored to the profile. "
1920 )
1921 user = (
1922     f"AGENT PROFILE:{profile}\n"
1923     f"TRUE CAR QUALITY: {omega}\n\n"
1924     f"CAR FACTS: {car_facts}\n"
1925     f"BASELINE MESSAGE:{baseline_message}}\n"
1926     f"BASELINE MESSAGE precision: {p}\n"
1927     f"DESIRED MESSAGE precision: {q}\n"
1928     f"IMPORTANT: Output a message with the requested precision. "
1929     f"TASK: Write a message (MAX 100 words) with the desired precision."
1930     f'JSON response: {"message": "<message>}}'
1931 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We now report further experiments that validate the LLMs' self-reported beliefs (shown in figure 1b) and the LLM oracle's performance. Here we treat the LLMs as rational, utility-maximizing—though not necessarily Bayesian—agents. These experiments have two goals: (1) to demonstrate that the model's self-reported posterior genuinely coincides with the belief governing its choices in state-contingent decision problems, and (2) to confirm that messages produced by the oracle indeed carry the requested precision.

First, we provide the LLM with a prior and a new signal, obtain its self-reported posterior belief, and then ask it to make a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent should choose “buy” exactly when its subjective belief exceeds 0.5. figure 4a shows the LLM's decisions snapping sharply to this threshold: it chooses “buy” overwhelmingly when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model's willingness to pay for a payoff contingent on the true state. For a range of prices c , we ask whether the agent would pay c in exchange for unit utility if the state is good. A rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs and shows a clean separation: acceptance rates are high when $c < p$ and low when $c > p$, exactly as a utility-maximizing decision maker with belief p requires. Both experiments confirm that the self-reported posterior faithfully tracks the belief the model actually uses in its decisions.

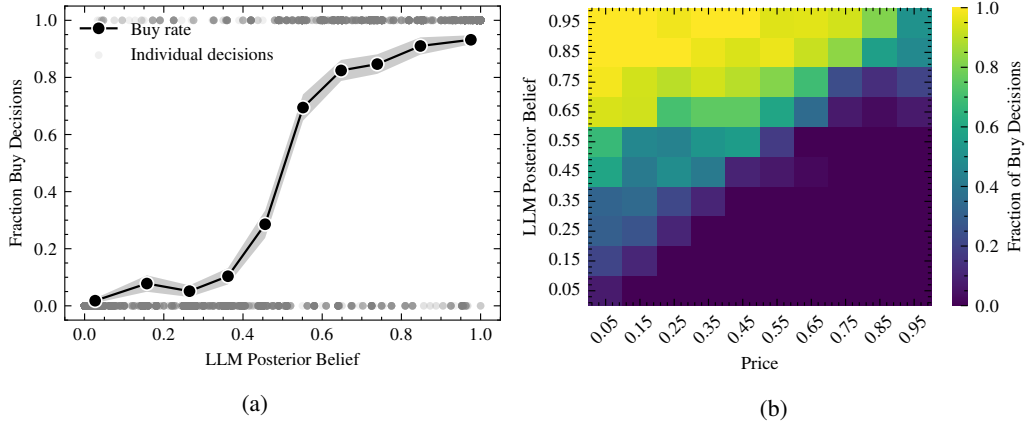


Figure 4: The rate at which the LLM chose a ‘buy’ decision as a function of (a) its self-reported posterior belief and (b) its self-reported posterior together with the price of buying. Both panels confirm that the reported belief drives the decision.

We likewise validate the oracle’s ability to craft messages of a given precision. After the oracle generates a message of specified precision q , we have a separate LLM estimate its precision $\hat{q}$. Running this for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ over 100 programmatically generated agent profiles with 10 trials per combination—6,000 trials in total—yields a mean absolute error $|q - \hat{q}|$ of just 0.025. The realized precision of the oracle’s messages therefore closely follows the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now unpack belief trajectories from an example run of the LLM experiment to make the impact of non-Bayesian belief updating concrete. We fix the policy to be the analytically optimal one and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, whereas the LLM belief (solid black line) is markedly more “stubborn.” This macro-level trajectory follows directly from the micro-level biases: the public belief resists entering a cascade (NB3) precisely because individual agents overreact to the occasional positive signal that cuts against the group’s increasingly negative belief (NB2).

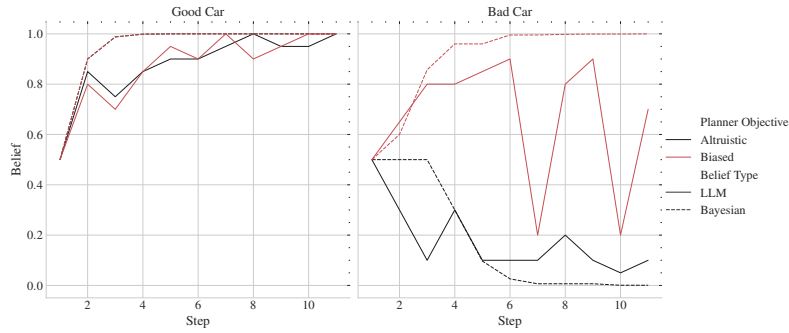


Figure 5: Agent belief-updating trajectories for the true Bayesian belief and the belief reported by the LLM agent under analytically optimal policies, across planner objectives and true states of the world. The persistent gap between the two makes the non-Bayesian “stubbornness” of the LLM agents plain.